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October 26, 2026

Professor Zidong Wang
Managing Editor-in-Chief
Neurocomputing

Dear Professor Wang,

We submit for your consideration the manuscript "When Do Kolmogorov–Arnold Networks Help in Wireless? A Matched-Capacity, Non-Saturated Benchmark" as a Regular Article in *Neurocomputing*. The work is authored by Phuc Hao Do (Danang Architecture University) and Minh Tuan Pham (The University of Danang, University of Science and Technology); I serve as corresponding author.

Kolmogorov–Arnold networks move the nonlinearity of a multilayer perceptron from the nodes onto learnable univariate functions on the edges, and are widely reported to match or beat MLPs at fewer parameters while remaining interpretable. The manuscript asks when that edge-wise basis actually helps, and why. It is a study of a learning architecture: three signal-processing tasks serve as the testbed that makes the question concrete and resolvable, and the object under study throughout is the network family rather than any one application.

The contribution is twofold. First, a parameter-matched, multi-seed protocol with paired confidence intervals, applied to four KAN instantiations against MLP, convolutional and classical baselines, which turns "KAN uses fewer parameters" into a testable statement rather than an assumption. Second, and the reason the result transfers beyond the three tasks, a controlled experiment that identifies the variable governing the outcome: the share of the target that is additive rather than carried by cross-coordinate interaction. On a synthetic family whose difficulty is held fixed by a structure-agnostic control at the same parameter budget, the KAN advantage decays monotonically with interaction, reverses sign in the small-sample regime, and erodes without reversing when data is plentiful. The ordering holds across all four instantiations and survives matching network depth on both sides.

The findings are two-sided by design, and several run against the architecture the manuscript examines as well as against our own earlier choices: a symmetric hyperparameter sweep widens the gap rather than closing it, one of our own baselines is shown to have been under-tuned and the correction is reported, and one earlier comparison is redone as a paired test with the interval it should always have carried. We believe an evaluation that reports what it finds, in both directions, is more useful to the community than another favourable benchmark.

The work is original, has not been published previously, and is not under consideration for publication elsewhere. The authors declare no competing financial or personal interests. This work was supported by Funds for Science and Technology Development of the University of Danang under Project B2025-DN02-26. The use of generative AI in preparing the manuscript is disclosed in a dedicated section before the reference list; no generative tool produced, altered, or interpreted any experimental measurement. All code, data-preparation scripts and result files that reproduce every reported number are openly available at

https://github.com/haodpsut/kan-wireless-benchmark, and the manuscript maps each table and figure to the script and result file that generate it.

Thank you for considering this submission.

Yours sincerely,

Phuc Hao Do
(on behalf of both authors)
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